A3 123090612

Question 1

1. Generating Variables and Natural Logs

- **Generated Variables**:
  - `yield` : yield $= \frac{\text{hx95_h125}}{\text{hx95_nh112}}$
  - `fertilizer` : fertilizer $= \frac{\text{nh6_3}}{\text{hx95_nh112}}$
  - `migrate` : A binary variable indicating whether the household had any migration days in 2001:

$$\text{migrate} = \begin{cases} 1 & \text{if h95_nh269} > 0 \\ 0 & \text{otherwise} \end{cases}$$

  - **Natural Logarithms**:
    - $\text{ln_yield} = \log(\text{yield} + 1)$
    - $\text{ln_fertilizer} = \log(\text{fertilizer} + 1)$
- **Interpretation**:
  - These transformations ensure that zeros in the data (e.g., no fertilizer used or no yield) are handled smoothly in the natural logarithm calculations.

```
* Set the directory for the datasets
cd "D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A3"

* Load the household data
use aghousehold.dta, clear

* Question 1: Limit the sample to households in the year 2001
keep if year == 2001

* Generate the variables 'yield' and 'fertilizer'
gen yield = hx95_nh125 / hx95_nh112
gen fertilizer = nh6_3 / hx95_nh112

* Generate a binary variable for migration decision
gen migrate = (h95_nh269 > 0)

* Generate natural log variables for yield and fertilizer
* (log + 1 transformation to smooth zero values)
gen ln_yield = log(yield + 1)
gen ln_fertilizer = log(fertilizer + 1)
```

```
. do "D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A3\Q1_4.do"

. * Set the directory for the datasets
. cd "D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A3"
D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A3

.
. * Load the household data
. use aghousehold.dta, clear

.
. * Question 1: Limit the sample to households in the year 2001
. keep if year == 2001
(38,832 observations deleted)

.
. * Generate the variables 'yield' and 'fertilizer'
. gen yield = hx95_nh125 / hx95_nh112
(1,268 missing values generated)

. gen fertilizer = nh6_3 / hx95_nh112
(1,268 missing values generated)

.
. * Generate a binary variable for migration decision
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.
. * Generate natural log variables for yield and fertilizer (log + 1 transformation to smooth zero value
> s)
. gen ln_yield = log(yield + 1)
(1,268 missing values generated)

. gen ln_fertilizer = log(fertilizer + 1)
(1,268 missing values generated)
```

2. Linear Probability Model of Migration Decision on `ln_yield`

- **Regression model**:

$$\text{migrate} = \beta_0 + \beta_1 \cdot \ln(\text{yield}) + \varepsilon$$

- **Results**:
  - Coefficient of `ln_yield` (β_1): **0.0495** (Std. Err.: **0.0128**)
  - Intercept (`_cons`): **0. 1773844** (Std. Err.: **0.743338**)
  - p -value: **0.000**
  - 95% confidence interval: [0.0244, 0.0747]
- **Interpretation**:
  The coefficient indicates that a 1% increase in yield (log-transformed) is associated with a **0.0495/100 increase in the probability of migration**. The result is statistically significant ($p < 0.05$).

```
.
. * Question 2: Run a linear probability model of household migration decision on ln_yield
. reg migrate ln_yield
```

Source	SS	df	MS		
Model	3.68868145	1	3.68868145	Number of obs	= 4,200
Residual	1040.44251	4,198	.247842427	F(1, 4198)	= 14.88
				Prob > F	= 0.0001
Total	1044.13119	4,199	.24866187	R-squared	= 0.0035
				Adj R-squared	= 0.0033
				Root MSE	= .49784

migrate	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
ln_yield	.0495398	.0128412	3.86	0.000	.0243642	.0747155
_cons	.1773844	.0743338	2.39	0.017	.0316509	.3231179

3. Potential Sources of Bias in the OLS Point Estimate

1. **Measurement Error in Yield**:
   - Yield is often misreported or inaccurately measured. This introduces classical measurement error, which biases the OLS estimate **downward**.
2. **Simultaneity:**
   - Migration might affect yield, causing endogeneity to be a potential bias.
   - $migrate_i = \beta_0 + \beta_1 yield_i + \epsilon_i$
   - $yield_i = \gamma_0 + \gamma_1 migrate_i + u_i$
   - $\hat{\beta}_1 = \beta_1 + \rho \frac{\sigma_\epsilon}{\sigma_{yield}} \quad where \rho = Corr(\epsilon, yield)$
   - If $\rho > 0$, the bias is **upward**
   - If $\rho < 0$, the bias is **downward**
3. **Omitted Variable Bias**:
   - Variables like household wealth might affect both yield and migration, causing **Upward bias**.

4. First Stage Regression: Instrument Relevance

- **First Stage Regression Model**:

$$\text{ln_yield} = \beta_0 + \beta_1 \cdot \text{ln_av_rain} + \varepsilon$$

- **Results**:
  - Coefficient of `ln_av_rain` (β_1): **0.4645** (Std. Err.: **0.0196**)
  - Intercept (`_cons`): **2.526537** (Std. Err.: **0.1365663**)
  - p -value: **0.0000**
  - 95% confidence interval: [0.4260, 0.5030]
  - R^2: **0.1229**
- **Interpretation**:
  - The coefficient suggests that a 1% increase in average rainfall (log-transformed) leads to a **0.4645% increase in agricultural yield (log-transformed)**.
  - The coefficient of `ln_av_rain` (0.4645) is positive and statistically significant ($p < 0.01$). This suggests that average rainfall (log-transformed) is a relevant predictor of agricultural yield (log-transformed).
  - The R-squared value (0.1229) indicates that about 12.3% of the variation in `ln_yield` is explained by `ln_av_rain`.

```
.
. * Question 4: First stage regression for IV relevance
. * Generate the natural log of average rainfall
. gen ln_av_rain = log(av_rain)
(448 missing values generated)

.
. * Make sure the 2SLS is operate on the same, valid dataset
. drop if missing(ln_yield, ln_av_rain, migrate)
(1,838 observations deleted)

.
. * Run the first stage regression: ln_yield on ln_av_rain
. reg ln_yield ln_av_rain
```

Source	SS	df	MS		
Model	182.285667	1	182.285667	Prob > F	0.0000
Residual	1300.86357	4,001	.325134609	R-squared	0.1229
Total	1483.14924	4,002	.370602008	Adj R-squared	0.1227
				Root MSE	.57021

Number of obs	=	4,003
F(1, 4001)	=	560.65

ln_yield	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
ln_av_rain	.4644991	.0196173	23.68	0.000	.4260382	.50296
_cons	2.526537	.1365663	18.50	0.000	2.258791	2.794284

5. Second Stage IV Regression (Manual Two-Stage Approach)

- **Second Stage Regression Model**:

$$\text{migrate} = \beta_0 + \beta_1 \cdot \text{ln_yield_hat} + \varepsilon$$

- **Results**:
  - Coefficient for `ln_yield_hat`: **0.4296** (Std. Err.: **-2.0128**)
  - Intercept (`_cons`): **-2.012801** (Std. Err.: **0.208899**)
  - p-value: **0.0000**
  - R-squared: **0.0338**
  - 95% confidence interval: [0.3584, 0.5007]
- **Interpretation**:
  - The coefficient of `ln_yield_hat` (0.4296) is positive and statistically significant ($p < 0.01$), suggesting a positive causal effect of agricultural yield on the probability of household migration.
  - The IV coefficient indicates that 1% increase in predicted yield (instrumented by rainfall) is associated with **0.4296/100 increase in the probability of migration**.
  - The R-squared value (0.0338) indicates that only 3.38% of the variation in migration decisions is explained by the instrumented yield variable, which is common in binary outcome models.

```
. * Question 5: Second stage IV regression (manually, two-step approach)
. * First step: Predict ln_yield using ln_av_rain
. predict ln_yield_hat, xb

.
. * Check whether all the data are valid
. summ ln_yield_hat
```

Variable	Obs	Mean	Std. dev.	Min	Max
ln_yield_hat	4,003	5.753103	.2134213	5.320964	6.0853

```
. list if missing(ln_yield_hat)

.
. * Second step: Use ln_yield_hat as an instrument for ln_yield in the migration regression
. reg migrate ln_yield_hat
```

Source	SS	df	MS		
Model	33.6398682	1	33.6398682	Number of obs	= 4,003
Residual	960.267701	4,001	.240006924	F(1, 4001)	= 140.16
				Prob > F	= 0.0000
				R-squared	= 0.0338
				Adj R-squared	= 0.0336
Total	993.907569	4,002	.248352716	Root MSE	= .48991

migrate	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
ln_yield_hat	.4295868	.0362857	11.84	0.000	.3584465	.500727
_cons	-2.012801	.208899	-9.64	0.000	-2.422359	-1.603243

```
.
. * Caompare to the first & second stage of the 2SLS
. count if !missing(ln_yield, ln_av_rain, migrate)
  4,003

. count if !missing(ln_yield_hat, migrate)
  4.003
```

6

```
.
. * Question 6: Use ivregress 2sls to directly estimate the IV model
. ivregress 2sls migrate (ln_yield = ln_av_rain)

Instrumental variables 2SLS regression          Number of obs   =      4,003
                                                Wald chi2(1)    =     111.36
                                                Prob > chi2     =     0.0000
                                                R-squared       =          .
                                                Root MSE        =     .54961

------------------------------------------------------------------------------
     migrate | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
    ln_yield |   .4295868   .0407081    10.55   0.000     .3498003    .5093732
       _cons |  -2.012801    .234359    -8.59   0.000    -2.472136   -1.553466
------------------------------------------------------------------------------
Endogenous: ln_yield
Exogenous:  ln_av_rain
```

(a) IV Estimation Using `ivregress 2sls`

- **Model**:

$$\text{migrate} = \beta_0 + \beta_1 \cdot \text{ln_yield} + \varepsilon$$

  with `ln_av_rain` as an instrument for `ln_yield`.
- **Results**:
  - Coefficient for `ln_yield`: **0.4296** (Std. Err.: **0.0407**)
  - Intercept (`_cons`): **-2.012801** (Std. Err.: **0.234359**)
  - z-value: **10.55**
  - p-value: **0.0000**
  - 95% confidence interval: [0.3498, 0.5094]
- **Comparison of Manual 2-Stage Regression and `ivregress` Results**:
  - Coefficient for `ln_yield_hat`:
    - Manual 2-stage: **0.4296**(Std. Err.: **-2.0128**)
    - `ivregress 2sls`: **0.4296**(Std. Err.: **0.0407**)
  - Intercept (`_cons`):
    - Manual 2-stage: **-2.012801** (Std. Err.: **0.208899**)
    - `ivregress 2sls`: **-2.012801** (Std. Err.: **0.234359**)
- **Conclusion**:
  - By using two different ways, we get the same results with no difference, but the standard error is relatively larger than we got in (5).
  - This is because in manual methods, the standard error of `ln_yield` does not be considered the variability introduced in the first stage regression.

(b) Comparison and Discussion

1. **OLS Estimate (0.0495) vs. IV Estimate (0.4296):**
    - The OLS estimate is much smaller than the IV estimate, which suggests that the OLS regression is biased.
    - Possible reasons for this bias include:
        - **Measurement error** in yield, which typically biases OLS estimates downward.
        - **Endogeneity**: Migration decisions might influence yield, leading to reverse causality.
        - **Omitted variable bias**: Unobserved factors such as household-specific risk preferences or weather conditions could affect both yield and migration.
2. **Direction of Bias**:
    - The IV estimate is larger than the OLS estimate, which is consistent with **downward** bias in OLS due to measurement error or reverse causality.
3. **Comparison Between Manual and `ivregress` Results**:
    - The results from the manual IV estimation and the `ivregress` command are very similar, confirming the robustness of the results. Slight differences in standard errors and coefficients reflect computational nuances.

The full do & log file

```
* Set the directory for the datasets
cd "D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A3"

* Load the household data
use aghousehold.dta, clear

* Question 1: Limit the sample to households in the year 2001
keep if year == 2001

* Generate the variables 'yield' and 'fertilizer'
gen yield = hx95_nh125 / hx95_nh112
gen fertilizer = nh6_3 / hx95_nh112

* Generate a binary variable for migration decision
gen migrate = (h95_nh269 > 0)

* Generate natural log variables for yield and fertilizer
* (log + 1 transformation to smooth zero values)
gen ln_yield = log(yield + 1)
gen ln_fertilizer = log(fertilizer + 1)

* Save the cleaned household data for merging
save cleaned_aghousehold.dta, replace

* Load the rainfall data
use village_rainfall.dta, clear

* Save the cleaned rainfall data
save cleaned_village_rainfall.dta, replace

* Merge the two datasets on village ID (vl_id)
use cleaned_aghousehold.dta, clear
merge m:1 vl_id using cleaned_village_rainfall.dta

* Question 2: Run a linear probability model of household migration decision on ln_yield
reg migrate ln_yield

* Question 4: First stage regression for IV relevance
* Generate the natural log of average rainfall
gen ln_av_rain = log(av_rain)

* Make sure the 2SLS is operate on the same, valid dataset
drop if missing(ln_yield, ln_av_rain, migrate)

* Run the first stage regression: ln_yield on ln_av_rain
reg ln_yield ln_av_rain

* Question 5: Second stage IV regression (manually, two-step approach)
* First step: Predict ln_yield using ln_av_rain
predict ln_yield_hat, xb

* Check whether all the data are valid
summ ln_yield_hat
list if missing(ln_yield_hat)

* Second step: Use ln_yield_hat as an instrument for ln_yield in the migration regression
reg migrate ln_yield_hat

* Caompare to the first & second stage of the 2SLS
count if !missing(ln_yield, ln_av_rain, migrate)
count if !missing(ln_yield_hat, migrate)

* Question 6: Use ivregress 2sls to directly estimate the IV model
```

```
63  * Question 6: Use ivregress 2sls to directly estimate the IV model
64  ivregress 2sls migrate (ln_yield = ln_av_rain)
```

```
------------------------------------------------------------------------------------------------------------
      name:  <unnamed>
       log:  D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A3\Q1_5.log
  log type:  text
 opened on:  26 Nov 2024, 17:54:09

. do "D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A3\Q1_5.do"

. * Set the directory for the datasets
. cd "D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A3"
D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A3

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. use aghousehold.dta, clear

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(38,832 observations deleted)

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. * (log + 1 transformation to smooth zero values)
. gen ln_yield = log(yield + 1)
(1,268 missing values generated)

. gen ln_fertilizer = log(fertilizer + 1)
(1,268 missing values generated)

.
. * Save the cleaned household data for merging
. save cleaned_aghousehold.dta, replace
(file cleaned_aghousehold.dta not found)
file cleaned_aghousehold.dta saved

.
. * Load the rainfall data
. use village_rainfall.dta, clear

.
. * Save the cleaned rainfall data
. save cleaned_village_rainfall.dta, replace
(file cleaned_village_rainfall.dta not found)
file cleaned_village_rainfall.dta saved

.
. * Merge the two datasets on village ID (vl_id)
. use cleaned_aghousehold.dta, clear

. merge m:1 vl_id using cleaned_village_rainfall.dta

    Result                      Number of obs
    -----------------------------------------
    Not matched                           373
        from master                         0  (_merge==1)
        from using                        373  (_merge==2)

    Matched                             5,468  (_merge==3)
    -----------------------------------------
```

```
.
. * Question 2: Run a linear probability model of household migration decision on ln_yield
. reg migrate ln_yield

      Source |       SS           df       MS      Number of obs   =     4,200
-------------+----------------------------------   F(1, 4198)      =     14.88
       Model |  3.68868145         1  3.68868145   Prob > F        =    0.0001
    Residual |  1040.44251     4,198  .247842427   R-squared       =    0.0035
-------------+----------------------------------   Adj R-squared   =    0.0033
       Total |  1044.13119     4,199   .24866187   Root MSE        =    .49784

------------------------------------------------------------------------------
     migrate | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    ln_yield |   .0495398   .0128412     3.86   0.000     .0243642    .0747155
       _cons |   .1773844   .0743338     2.39   0.017     .0316509    .3231179
------------------------------------------------------------------------------

.
. * Question 4: First stage regression for IV relevance
. * Generate the natural log of average rainfall
. gen ln_av_rain = log(av_rain)
(448 missing values generated)

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. * Make sure the 2SLS is operate on the same, valid dataset
. drop if missing(ln_yield, ln_av_rain, migrate)
(1,838 observations deleted)

.
. * Run the first stage regression: ln_yield on ln_av_rain
. reg ln_yield ln_av_rain

      Source |       SS           df       MS      Number of obs   =     4,003
-------------+----------------------------------   F(1, 4001)      =    560.65
       Model |  182.285667         1  182.285667   Prob > F        =    0.0000
    Residual |  1300.86357     4,001  .325134609   R-squared       =    0.1229
-------------+----------------------------------   Adj R-squared   =    0.1227
       Total |  1483.14924     4,002  .370602008   Root MSE        =    .57021

------------------------------------------------------------------------------
    ln_yield | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
  ln_av_rain |   .4644991   .0196173    23.68   0.000     .4260382      .50296
       _cons |   2.526537   .1365663    18.50   0.000     2.258791    2.794284
------------------------------------------------------------------------------

.
. * Question 5: Second stage IV regression (manually, two-step approach)
. * First step: Predict ln_yield using ln_av_rain
. predict ln_yield_hat, xb

.
. * Check whether all the data are valid
. summ ln_yield_hat

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
ln_yield_hat |      4,003    5.753103    .2134213   5.320964     6.0853

. list if missing(ln_yield_hat)

.
. * Second step: Use ln_yield_hat as an instrument for ln_yield in the migration regression
. reg migrate ln_yield_hat

      Source |       SS           df       MS      Number of obs   =     4,003
-------------+----------------------------------   F(1, 4001)      =    140.16
       Model |  33.6398682         1  33.6398682   Prob > F        =    0.0000
    Residual |  960.267701     4,001  .240006924   R-squared       =    0.0338
-------------+----------------------------------   Adj R-squared   =    0.0336
       Total |  993.907569     4,002  .248352716   Root MSE        =    .48991

------------------------------------------------------------------------------
     migrate | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
ln_yield_hat |   .4295868   .0362857    11.84   0.000     .3584465     .500727
```

```
       _cons |  -2.012801    .208899    -9.64   0.000    -2.422359   -1.603243
------------------------------------------------------------------------------

.
. * Caompare to the first & second stage of the 2SLS
. count if !missing(ln_yield, ln_av_rain, migrate)
  4,003

. count if !missing(ln_yield_hat, migrate)
  4,003

.
. * Question 6: Use ivregress 2sls to directly estimate the IV model
. ivregress 2sls migrate (ln_yield = ln_av_rain)

Instrumental variables 2SLS regression          Number of obs   =      4,003
                                                Wald chi2(1)    =     111.36
                                                Prob > chi2     =     0.0000
                                                R-squared       =          .
                                                Root MSE        =     .54961

------------------------------------------------------------------------------
     migrate | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
    ln_yield |   .4295868   .0407081    10.55   0.000     .3498003    .5093732
       _cons |  -2.012801    .234359    -8.59   0.000    -2.472136   -1.553466
------------------------------------------------------------------------------
Endogenous: ln_yield
Exogenous:  ln_av_rain

.
end of do-file

. log close
      name:  <unnamed>
       log:  D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A3\Q1_5.log
  log type:  text
 closed on:  26 Nov 2024, 17:54:26
-------------------------------------------------------------------------------------------------------------------
```

Question 2

1. Expected Signs of β_0 and β_1

- **Sign of** β_0: We expect $\beta_0 > 0$. Fertilizer is a key input to crop production, so an increase in fertilizer usage is likely to result in a higher crop yield, holding other factors constant.
- **Sign of** β_1: We also expect $\beta_1 > 0$. Farmers are likely to use more fertilizer in areas with historically higher crop yields, as higher yields may signal more fertile land or favorable conditions, incentivizing higher fertilizer application.

2. Bias in the OLS Estimator for β_0 and β_1

The OLS estimators for β_0 and β_1 are biased because of **simultaneity bias**. The two equations are simultaneous, meaning that the dependent variable in one equation (Crop or Fertilizer) appears as an independent variable in the other. This creates endogeneity due to reverse causality.

Bias Directions

By solving two equations:

$$\begin{cases} Crop = \alpha_0 + \beta_0 Fertilizer + u & (1) \\ Fertilizer = \alpha_1 + \beta_1 Crop + v & (2) \end{cases}$$

We have:

$$Crop = \frac{\alpha_0 + \beta_0\alpha_1 + \beta_0 v + u}{1 - \beta_0\beta_1}$$

$$\hat{\beta}_0 = \beta_0 + \frac{Cov(Fertilizer, u)}{Var(Fertilizer)}$$

$$\because \quad E[Fertilizer|u] \neq 0, \quad Cov(Fertilizer, u) > 0$$

$$\therefore \quad \text{OLS of} \quad \hat{\beta}_0 \quad \text{will larger than} \quad \beta_0, \quad \text{which means Upward}$$

Based on the same calculation, we can also get the expression of $\hat{\beta}_1$:

$$\hat{\beta}_1 = \beta_1 + \frac{Cov(Crop, v)}{Var(Crop)}$$

$$\because \quad E[Crop|v] \neq 0, \quad Cov(Crop, v) > 0$$

$$\therefore \quad \text{OLS of} \quad \hat{\beta}_1 \quad \text{will larger than} \quad \beta_1, \quad \text{which means Upward}$$

3. Two-Stage Least Squares (2SLS) and Instrument Selection

To resolve the simultaneity problem, we use **instrumental variables (IV)** for Fertilizer and Crop. An effective instrument must satisfy two conditions:

1. **Relevance**: The instrument must be correlated with the endogenous variable it is replacing (Fertilizer or Crop).
2. **Exogeneity**: The instrument must not be correlated with the error term in the equation being estimated.

Instruments for Independent Variable Crop, Dependent Variable Fertilizer

- **Sunshine**: Sunshine is likely to affect crop yields directly but is unlikely to influence the error term in the Fertilizer equation. It satisfies exogeneity because it is determined by natural conditions and does not depend on farmer decisions.

$$Cov(Sunshine, Crop) \neq 0, \quad Cov(Sunshine, v) = 0$$

Instruments for Independent Variable Fertilizer, Dependent Variable Crop

- **Budget**: The farmer's budget affects how much fertilizer they can afford to apply but is unlikely to directly influence crop yield (beyond its effect on Fertilizer usage). It satisfies exogeneity because it is not directly determined by crop production outcomes.

$$Cov(Budget, Fertilizer) \neq 0, \quad Cov(Budget, u) = 0$$